TOOL CATALOGUE AND TOOL/API SCHEMAS

BSE4104 EMERGING TRENDS IN SOFTWARE ENGINEERING

University Student-Support Case Agent

1. Purpose

This document defines the tools (functions) the University Student-Support Case Agent may call at inference time to answer freshman onboarding questions and, where a question can't be resolved from available information, raise a support case on the student's behalf. Each tool below is either a read-only retrieval action or a low-risk simulated side effect that stores a draft the agent cannot withdraw - the agent never takes an action that resolves, decides, or otherwise carries a high-impact outcome; those actions remain outside its authority per the Project Charter.

Two tools are defined, consistent with the Project Charter's Minimum Proposal Statement, which scopes the agent to "a document-retrieval (RAG) tool and a case-creation tool." retrieve_documents retrieves grounded passages - from the approved corpus by calling retrieve() in src/rag/retrieve.py - so the agent can answer with a citation. On the tool-enabled path, this call replaces the automatic context injection in llm_client.py. Prompt v1.2 keeps that automatic injection and does not call this tool. create_support_case performs a low-risk simulated side effect, storing a local draft and a queue label for an issue the corpus cannot resolve. It sends no message to an office. A person later reads the stored draft.

2. Tool Catalogue -Overview

Tool name	Purpose	Type	Maps to
retrieve_documents	Retrieve grounded passages from the freshman onboarding corpus - using only the Corpus/Source Register - to answer a question	Read-only retrieval	US-1, US-2, US-3
create_support_case	Store a local draft and a queue label for an issue the corpus cannot resolve	Simulated write / side effect	US-4, US-5, US-10

3. Tool Schemas

3.1 retrieve_documents /knowledge base.

Purpose: Given a student's question, retrieve the most relevant passages from the approved freshman-onboarding corpus (Corpus/Source Register) - using only the Corpus/Source Register - so the agent can construct a grounded, cited answer instead of relying on model memory. The tool calls retrieve() in src/rag/retrieve.py, returns at most 3 chunks, and drops every chunk below similarity 0.70. On the tool-enabled path the application does not inject context before the model call; the model must call this tool. v1.2 still injects context automatically and does not call this tool.

Input schema. The model may supply only the properties in this object.

```
{
  "type": "object",
  "required": ["query"],
  "properties": {
    "query": {
```

```
      "type": "string",
      "minLength": 1,
      "description": "The student's question or a reformulated search query."
    },
    "top_k": {
      "type": "integer",
      "minimum": 1,
      "maximum": 3,
      "description": "Number of chunks to return. Optional. When omitted, the tool uses 3.
Chunks below similarity 0.70 are omitted."
    }
  }
}
```

Output schema. The tool implementation sets every property below. These fields are not model arguments.

```
{
  "type": "object",
  "required": ["status", "chunks", "retrieved_count"],
  "properties": {
    "status": {
      "type": "string",
      "enum": ["ok", "no_match", "error"]
    },
    "chunks": {
      "type": "array",
      "items": {
        "type": "object",
        "required": ["chunk_id", "text", "source_title", "source_url",
"similarity_score"],
        "properties": {
          "chunk_id": { "type": "string" },
          "text": {
            "type": "string",
            "description": "The retrieved passage."
          },
          "source_title": { "type": "string" },
          "source_url": { "type": "string" },
          "section": {
            "type": "string",
            "description": "Optional heading within the source. Omit when the chunk has
none."
          },
          "similarity_score": {
            "type": "number",
            "minimum": 0,
            "maximum": 1
          }
        }
      }
    },
    "retrieved_count": {
      "type": "integer",
      "minimum": 0
    },
    "message": {
      "type": "string",
      "description": "Present when status is error. Example: query is required. Omit when
status is ok or no_match."
    }
```

```
  }
}
```

Authorization: None beyond standard application access - every freshman using the agent may invoke this tool. It is read-only against a public, pre-approved corpus (see Corpus/Source Register); it cannot read or write student, case, or ticket data. The agent must not use retrieved text to answer an admissions, grading, or disciplinary matter. It declines and directs the student to the relevant office, and it does not open a case for that matter.

Failure behaviour

- Missing query: return status "error" with message "query is required"; the agent must not guess a query and must ask the student to rephrase.
- No relevant chunks at or above similarity 0.70: return status "no_match" with an empty chunks array; the agent must use the honest-fallback template (US-2), never fabricate a source.
- Vector index unavailable (e.g. index file missing or service down): return status "error" with a generic message; the agent must use the Failure Behaviour template from Prompt Specification v1.1 ("Something went wrong on my end...") and must not answer from general knowledge as a substitute.
- Unexpected/malformed tool response (e.g. missing source_url on a chunk): the agent must discard that chunk rather than cite it without a source, and proceed with whatever valid chunks remain. If no valid chunk remains, return status "no_match".
- A requested fact is not in the corpus, including a live timetable entry or a registration deadline: return status "no_match". The agent must use the honest-fallback template and must not guess a date or class time. It may offer to open a support case, and it may call create_support_case only after the student has agreed.

3.2 create_support_case

Purpose: Draft a support case record when a student's issue cannot be resolved from the corpus (US-4), so the application stores a local draft and a queue label from the category (US-5). It sends no message to an office, and the agent cannot withdraw, resolve, or close the draft. Per the AI Boundary Matrix, opening that draft is a low-risk simulated side effect and does not need staff approval. Writing the record, validating inputs, and choosing the queue label are deterministic, not AI judgment. A person attends to the stored draft outside this system.

Input schema. The model may supply only the properties in this object. Student agreement and session_id are not properties here.

```
{
  "type": "object",
  "required": ["student_identifier", "issue_summary", "category"],
  "properties": {
    "student_identifier": {
      "type": "string",
      "pattern": "^\\d{2}/[UX]/\\d{4,5}(/(PS|EVE))?$",
      "description": "Registration number, compared after letters are uppercased. Shape:
two digits, a slash, U or X, a slash, four or five digits, and optionally a slash plus
only PS or EVE. Accepted examples: 23/U/93939, 23/U/9393, 23/U/93939/PS, 23/U/93939/EVE,
23/X/93939."
    },
    "issue_summary": {
      "type": "string",
      "minLength": 1,
```

```
      "description": "Short description of the unresolved issue."
    },
    "category": {
      "type": "string",
      "enum": [
        "it_webmail_wifi",
        "registration",
        "fees_informational",
        "accommodation",
        "other_unclassified"
      ]
    }
  }
}
```

Category-to-queue map

Application code chooses the queue label. The model cannot choose or change it. These are the only labels.

- it_webmail_wifi stores the label ict_helpdesk.
- registration stores the label registration_office.
- fees_informational stores the label fees_information. The draft may ask about a fee process. It must not state a fee decision.
- accommodation stores the label accommodation_office.
- other_unclassified stores the label general_onboarding. Admissions, grading, and disciplinary matters must not use this label.
- If one of these categories has no row in this map, return status error, omit routed_queue, and store the draft with the label manual_review.

Output schema. Application code alone sets status, case_id, session_id, routed_queue, created_at, and validation_errors. The model must not send them.

```
{
  "type": "object",
  "required": ["status"],
  "properties": {
    "status": {
      "type": "string",
      "enum": ["created", "validation_error", "error"]
    },
    "case_id": {
      "type": "string",
      "description": "Present only when status is created. Assigned by application code.
Example shape: CASE-2026-000482."
    },
    "session_id": {
      "type": "string",
      "description": "Present when status is created. Assigned by application code from
the id it created when this chat session started. The model must not supply it."
    },
    "routed_queue": {
      "type": "string",
      "description": "Queue label stored on the draft. One of ict_helpdesk,
registration_office, fees_information, accommodation_office, general_onboarding, or
manual_review. Chosen from the category-to-queue map. No message is sent to an office.
Assigned by application code."
    },
    "created_at": {
      "type": "string",
```

```
      "description": "ISO 8601 timestamp assigned by application code."
    },
    "validation_errors": {
      "type": "array",
      "description": "Present when status is validation_error.",
      "items": {
        "type": "object",
        "required": ["field", "message"],
        "properties": {
          "field": { "type": "string" },
          "message": { "type": "string" }
        }
      }
    }
  }
}
```

Authorization: The application records agreement only when the student explicitly confirms in the chat that a case should be opened (US-4). Until that confirmation is stored for the current session, create_support_case is left off the tool list sent to the model. A call that arrives without that confirmation is rejected by the orchestration layer before any handler runs, and no case is written. Agreement is not a model argument, because the model could mark a case confirmed without the student having agreed. The model may never call a “resolve”, “close”, or “reassign” action, because no such tool is exposed. Field validation and the category-to-queue lookup are performed by deterministic code, not the model. Admissions, grading, and disciplinary matters are not categories and must not be filed, including under other_unclassified. The orchestration layer rejects that call before a draft is written. fees_informational may record a question about a fee process. It must not record or imply a fee decision.

Failure behaviour

- Missing student_identifier or issue_summary: return status "validation_error" listing the missing field(s); the agent must prompt the student for the missing information before retrying (US-10), and must not fabricate a placeholder identifier.
- student_identifier is checked after letters are uppercased. It must be two digits, a slash, U or X, a slash, four or five digits, and optionally a slash followed only by PS or EVE, as in 23/U/93939, 23/U/9393, 23/U/93939/PS, 23/U/93939/EVE, and 23/X/93939. Any other shape, such as 23/A/93939 or 23/U/93939/DAY, returns status "validation_error"; the agent must ask the student to verify and correct it, not guess a corrected value.
- A category outside the enum returns status "validation_error" and no draft is written. A category inside the enum with no queue mapping returns status "error" with routed_queue omitted, and the draft is stored for manual review rather than the agent choosing an office, per US-5.
- Call before the student has agreed: the orchestration layer rejects the call before any handler runs. No case record is written and no case_id is issued. The agent must ask the student whether to open a case, and may try again only on a later turn after the student has explicitly agreed.
- Model supplies a session_id: the orchestration layer discards that value. The id stored on the case is the one the application created for this chat. If the application has no session id, the call is rejected before the handler and no case_id is issued.
- Unauthorized call (e.g. the model attempts to invoke a resolve/close action that is not in its tool list): the orchestration layer must reject the call before it reaches any handler, since no such tool is registered - this is enforced by the tool allow-list, not by prompting.

- Admissions, grading, or disciplinary request: the agent declines and directs the student to the relevant office. It does not call create_support_case. If a call still arrives, the orchestration layer rejects it before the handler. No draft is written and no case_id is issued.
- Storage/service unavailable: return status "error"; the agent must tell the student the case could not be created and to try again or contact the office directly, never claim success without a case_id.

4. Failure and Authorization Test Matrix

These are the minimum test conditions each tool must be exercised against, per the assignment's Week 4 activity to “test missing parameters, unauthorized requests, unavailable services and unexpected tool responses.” Evidence of these runs belongs in evidence/traces/ alongside the working demonstration.

Tool	Test condition	Expected result
retrieve_documents	Empty/missing query	status: error, no retrieval attempted
retrieve_documents	Query with no matching chunks	status: no_match, agent uses honest-fallback template
retrieve_documents	Vector index unreachable	status: error, agent uses generic failure message, no fabricated answer
retrieve_documents	Chunk missing source_title or source_url	Chunk dropped. If none remain, status no_match
create_support_case	Missing student_identifier	status: validation_error, agent re-prompts, no case_id issued
create_support_case	Malformed student_identifier	status: validation_error, agent asks student to correct the shape. Example invalid value: 23/A/93939
create_support_case	category outside the enum	status validation_error, no draft written
create_support_case	Attempted call to an unregistered resolve/close action	Rejected by tool allow-list before reaching any handler
create_support_case	Call before the student has agreed	Rejected before any handler runs, no case_id issued
create_support_case	Model supplies a session_id	Discarded. Stored id is the application session id
create_support_case	Admissions, grading, or disciplinary request	Declined and redirected. No draft, no case_id
create_support_case	Storage service unavailable	status: error, agent reports failure honestly, no case_id fabricated